W-D-661E
June 14, 1995
SUPERSEDING
W-D-661D
October 13, 1969

FEDERAL SPECIFICATION

DRILLS, ELECTRIC, PORTABLE (EXCLUSIVE OF HIGH FREQUENCY TYPES) AND STAND, PORTABLE ELECTRIC DRILL, VERTICAL

This specification is approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers portable electric drills, either 3-wire grounded or double insulated, and vertical drill stands with drills.

1.1.1 Federal specification coverage. Federal specifications do not cover all varieties of the commodity indicated by the title of the specification, or which are commercially available, but are intended to cover only those generally used by the Federal Government.

1.2 Classification.

1.2.1 Types, classes, and styles. The portable electric drills and drill stands shall be of the following types, classes, and styles, as specified (see 6.2):

Type I - High-speed drills, standard duty
Type II - Low speed drills, heavy duty
Type III - Medium speed drills

Class A - Heavy duty
Class B - Standard duty

Style 1 - Straight drive
Style 2 - 90° angle drive
Style 3 - 55° angle drive

Type IV - Drill stands with drills

2. APPLICABLE DOCUMENTS

2.1 Government documents. The following documents, of the issues in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

Federal Specification:

PPP-P-40 - Preservation and Packing of Hand Tools and Tool Accessories for Power Driven, Metal and Woodworking Machinery

Federal Standard:

FED-STD-H28 - Screw Threads Standards for Federal Services
FED-STD-123 - Marking for Domestic Shipment (Civilian Agencies)

FSC 5130

DISTRIBUTION STATEMENT A. Approved for public distribution; distribution is unlimited

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:33Z
Check the source to verify that this is the current version before use.

lot size of all components shall be 5000 [illegible] MIL-STD-105

Commercial Item Description:

A-A-51140 - Chucks, Drill, Key Type, Three-jaw, Geared

(Activities outside the Federal Government may obtain copies of Federal specifications, standards, and Commercial Item Descriptions as outlined under General Information in the Index of Federal Specifications, Standards and Commercial Item Descriptions. The Index, which includes cumulative bimonthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.)

(Single copies of this specification, and other Federal specifications and Commercial Item Descriptions required by activities outside the Federal Government for bidding purposes are available without charge from General Services Administration Business Service Centers in Boston, MA; New York, NY; Philadelphia, PA; Washington, DC; Atlanta, GA; Chicago, IL; Kansas City, MO; Fort Worth, TX; Houston, TX; Denver, CO; San Francisco, CA; Los Angeles, CA; and Seattle, WA.)

(Federal Government activities may obtain copies of Federal standardization documents and the Index of Federal Specifications, Standards and Commercial Item Descriptions from established distribution points in their agencies.)

Military Specification:

MIL-V-173 - Varnish, Moisture- and Fungus-Resistant (for the Treatment of Communications, Electronics, and Associated Electrical Equipment)

Military Standards:

MIL-STD-129 - Marking for Shipment and Storage
MIL-STD-461 - Electromagnetic Emission and Susceptibility Requirement for the Control of Electromagnetic Interference

(Copies of military specifications and standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Non-Government publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on the date of invitation for bids or request for proposal shall apply.

American National Standard Institute (ANSI), Inc.:

B5.10 - Machine Tapers

American National Standard Institute (ANSI)/Anti-Friction Bearing Manufacturers Association (AFBMA):

ANSI/AFBMA B1 - Terminology or Nomenclature for Various Types of Bearings and Parts Definitions and Illustrations

ANSI/AFBMA B10 - Balls: Terminology, Definitions and Tolerances for Metal Balls

ANSI/AFBMA B18.2 - Needle Roller Bearings, Radial, Inch Design Identification Code, Boundary Dimensions, Tolerances, Fitting and Mounting Practices

(Application for copies should be addressed to the American National Standards Institute, 11 West 42nd Street, New York, NY 10036.)

Source: https://assist.dla.mil -- Downloaded: 202610-26T11:33Z
Check the source to verify that this is the current version before use.

American National Standards Institute (ANSI)/American Society of Quality Control (ASQC):

ANSI/ASQC Z1.4 - Sampling Procedures and Tables for Inspection by Attributes

(Application for copies should be addressed to the American Society for Quality Control, P. O. Box 3005 611 E. Wisconsin Avenue, Milwaukee, WI 53201-4606.)

Underwriters' Laboratories, Inc. (UL):

UL No. 45 - Electric Tools

(Application for copies should be addressed to the Underwriters' Laboratories, Inc., 333 Pfingsten Road, Northbrook, IL 60062-2096.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

3. REQUIREMENTS

3.1 Illustrations. Illustrations herein are descriptive and not restrictive and are not intended to preclude the purchase of drills otherwise conforming to this specification.

3.2 First article inspection. When specified (see 6.2), the contractor shall furnish a completely fabricated portable electric drill for first article inspection in accordance with section 4. The sample shall be a preproduction item or a production item which meets the requirements of this specification. In either case, the sample shall be identical with the proposed items and shall be made or shall have been made by the methods that are to be used in production of items to fulfill the terms of the contract. Approval of the first article shall not relieve the contractor of the responsibility to furnish equipment in accordance with the requirements of this specification.

3.3 Material. All materials used in the fabrication of electric drills and drill stands shall be new and of a quality necessary to produce portable electric drills and drill stands to meet the requirements specified herein.

3.4 Design. The design shall be such as to provide convenient and safe operation. When specified, (see 6.2), the drill shall be of the double insulated design. All parts of the drill shall be designed and constructed to the requirements listed herein. All parts subject to wear shall be easily accessible for adjustment, replacement, or repair.

3.4.1 Conformance to Underwriters' Laboratories (UL), Inc. standards. The drill shall comply with the requirements of the specified Underwriters' Laboratories (UL) standards. The contractor shall submit proof to the procuring activity that the drill complies to the requirements of the applicable UL standards. Compliance shall be indicated by the permanent attachment of an appropriate label, seal, or stamp that contains the UL listing mark to the product, or by a certificate furnished therewith. In lieu of the aforementioned, the contractor shall submit independent proof, satisfactory to the Government, that the drill complies to the requirements of the applicable UL standards including the UL method of test. Contractors compliance with the UL requirements does not absolve the contractor from complete compliance with the requirements of this specification. In the event of difference between any of the requirements of this specification and those of UL, the requirements of this specification shall govern. When electromagnetic interference (EMI) suppression is required, only the basic drill, prior to modification for EMI suppression, shall comply to the requirements of the applicable UL standard.

3.4.2 Lubrication. The bearings, gears, and moving parts shall be assembled with lubricating medium. If lubricant replenishment is required, external lubricant fittings or other suitable means shall be provided. Sealed-for-life bearings and permanently lubricated gear boxes shall not be required to have external fittings. The gears shall operate in a lubricant which shall be prevented from excessive leakage from the gear casing or entering the motor enclosure. Parts requiring protection from dust shall be enclosed.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:33Z
Check the source to verify that this is the current version before use.

3.4.4 Interchangeability. All parts having the same manufacturer's number shall be manufactured to such standards that will permit replacement or adjustment without modification of the electric drill or its parts.

3.5 Construction. The drill shall be complete so that when connected to the specified power source it can be used for its designed operation. Construction shall be free of any characteristics or defects that will prevent the drill from passing any of the examinations and tests specified in section 4.

3.5.1 Castings. All castings shall be free from defects, scale, or mismatching. In no event shall such process as welding, peening, plugging, or filling with solders of metallic pastes be used in castings for reclaiming any part.

3.5.2 Surfaces. The surfaces of all castings, forgings, and molded parts shall be cleaned and free of sand, dirt, fins, spurs, scale, flux, or either harmful or extraneous materials. The external surfaces shall be smooth and all edges shall be rounded or beveled. The surfaces shall be finished according to the manufacturer's standard practice.

3.5.3 Welding, brazing, or soldering. Welding, brazing, or soldering shall be employed only where those operations are included in the fabrication of the original design. These operations shall not be employed as repair measures for defective parts.

3.5.4 Fastening devices. All screws, pins, bolts, and similar parts shall be installed with a means of preventing loss of tightness. All such parts when subject to removal or adjustment shall not be permanently deformed. All threaded parts shall be in the inch system and shall conform to FED-STD-H28.

3.6 Components. The drills furnished under this specification shall consist of a motor, switch, cable assembly, wiring, housing, gears and shafts, bearings, and handles. The drills and motors shall conform to applicable requirements and tests of UL 45.

3.6.1 Motor. The motors shall be universal, series wound or compensated series wound designed to operate on either 120 volt (± 10 percent) direct current (DC), or 120 volt (± 6 percent), single-phase, 25-60 cycle, alternating current (AC). The motor shall conform to the requirements of UL 45.

3.6.1.1 Commutator. The commutator segments shall be of hard drawn, hard rolled, or drop forged copper. The commutator shall have sheet mica separators between segments or shall be of the molded mica construction using material that will be immune to deformation and dielectric degradation when subjected to maximum rotational forces operating temperatures encountered in service.

3.6.1.2 Rotors and fans. The rotors, (armatures) and fans shall be balanced, dynamically, to insure freedom from destructive vibration.

3.6.2 Switch. Each drill shall be provided with a conveniently located switch for turning the current of the motor "OFF" and "ON". The inclusion of a locking pin or similar means for holding the switch in the "ON" position shall be at the option of the contractor, except for type III, class B, style 2 or 3 drills which shall be furnished with nonlocking switches with an instant release. All switches shall be double pole, except type I and type III, class B; 1/4 inch size drills may be furnished with single pole switches. The switches shall be rated for both AC & DC operations, except switches for drills sizes 1/2 inch or less which may be rated for either AC only operations or both AC & DC. The switches, except those which release automatically, shall be marked plainly to show the "OFF" and "ON" position. The switches shall comply with the requirements of UL 45. All reversible drills shall be supplied with a reversing device.

3.6.3 Wiring, cord, and plug cap. The wiring, flexible cord, and attachment plug cap shall be in accordance with UL 45, as applicable. The cord shall have an attachment plug cap and unless otherwise specified (see 6.2), shall have a minimum free length of 8 feet measured from point of entry into the drill. The attachment plug cap shall comply with UL 45 requirements for two-pole, three-wire grounding devices rated 15 amperes at 125 volts.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:33Z
Check the source to verify that this is the current version before use.

3.6.4 Housing. The housings shall be of a material of suitable characteristics to withstand stresses and mechanical shock imposed by normal usage. The housing shall include the motor case housing, gear case housing, and a handle housing.

3.6.5 Gears and shaft. The gears and shafts shall be manufactured from a suitably hardened alloy steel. All gears and shafts shall be supported by at least two bearings and shall be of such dimension as to prevent deflection. The gears shall have machine cut or roll formed teeth and be designed to efficiently transmit imposed loads. The gears and shafts, except the chuck end of the spindle shaft, shall be fully enclosed.

3.6.6 Bearings. All rotating parts of the drills shall be provided with precision quality commercially available ball and roller bearings conforming to ANSI/AFBMA B1, B10, and B18.2. Integral or separate shielding to prevent leakage of lubricant and entrance of contaminants shall be provided. Steel inserts may be omitted in mounting of bearings when a sufficient contact area and suitable fit between bearing outer-diameter and housing is provided.

3.6.7 Armature shaft bearings. The armature shaft bearings shall be of the ball or roller type. The ball, roller, or a combination of ball or roller and needle roller bearings may be used on drill armatures providing hardness of shaft, provisions for thrust loading and shielding are adequate. The type III, class B, size 3/8 or iess drills may use sleeve bearings at the commutator end of the armature shaft or in nonthrust positions.

3.6.8 Spindle bearings. The spindle arbor bearings shall be of either ball, roller, taper roller, or needle roller. If roller or needle roller bearings are used, axial thrust loads applied to the spindle shaft must be provided for by use of a separator thrust bearing, either ball or roller or an integral combination type roller thrust bearing.

3.6.9 Chucks and taper sockets. Unless otherwise specified (see 6.2), all chucks shall be in accordance with A-A-51140 and all taper sockets shall be in accordance with ANSI B5.10. All drills with chucks shall be furnished with a key and a thong. Type I and type III, class B, styles 2 and 3 drills shall be supplied with class 2, style A chucks. Type II and all other type III drills having a 1/2 inch capacity or less shall be supplied with class 1 or class 2, style A chucks. Drills having a 3/4 inch capacity shall have a drive end as specified (see 6.2). When specified (see 6.2), drills having a 3/4 inch capacity shall be supplied with a class 1, style B chuck with a matching arbor. Drills having a 1.0 inch or 1-1/4 inch capacity shall have an ANSI (Morse) No. 3 taper socket. On reversible drills, a means shall be provided to prevent loosening of the chuck.

3.7 Types.

3.7.1 Type I drills. Type I drills shall be high speed drills suitable for drilling aluminum or nonferrous metals of equivalent hardness. The drills shall conform to Table I for the size specified (see 6.2).

Table I. Type I drills, high speed, standard duty

Size (inch)	Chuck capacity (min.) (inch)	Weight without cord (max.) (pounds)	Length overall (max.) (inches)	Spindle revolutions per minute		Rated load amps at 120 volts AC (min.)	Torque at spindle shaft (min.)	
				No load (max.)	Rated load 1/		Rated load (foot-pounds)	Stall (foot-pounds)
1/4	0 - 1/4	5.5	12	4300	1500 - 3000	1.8	0.25	1.25

1/ The rated load speed shall in no case be less than 55 percent of the rated no-load speed.

3.7.1.1 Handles. The type I drills shall have an end handle with a pistol grip and trigger switch similar to Figure 1 and, when specified (see 6.2), shall have an end handle with a closed grip similar to Figure 2.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:33Z
Check the source to verify that this is the current version before use.

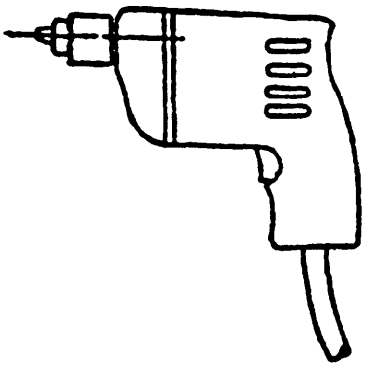


Figure 1. End handle pistol grip (1/4 inch and 3/8 inch capacity drills).

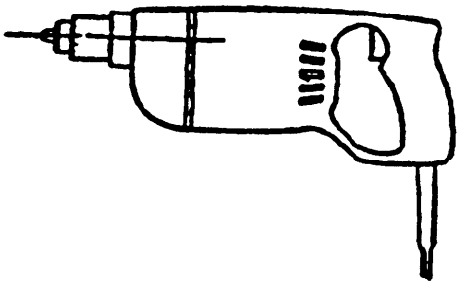


Figure 2. End handle closed grip (1/4 inch and and 3/8 inch capacity drills).

3.7.2 Type II drills. The type II drills shall be heavy duty low speed drills suitable for drilling-corrosion resistant steels. The type II drills shall conform to the requirements of Table II for the size specified.

Table II. Type II drills, low-speed, heavy duty

Size (inch)	Chuck capacity (min.) (inch)	Weight without cord (max.) (pounds)	Length overall (max.) (inches)	Spindle revolutions per minute		Rated load amps at 120 volts AC (min.)	Torque at spindle shaft (min.)	
				No load (max.)	Rated load 1/		Rated load (foot-pounds)	Stall (foot-pounds)
1/4	0 - 1/4	8.5	14	750	350 - 500	2.5	1.7	8.5
1/2	5/64 - 1/2	21	20	500	200 - 500	5.0	6.0	30.0

1/ The rated load speed shall in no case be less than 55 percent of the rated no-load speed.

3.7.2.1 Handles. Drills of 1/4 inch capacity shall have a pistol grip handle similar to Figures 1 or 3, or end handle similar to Figure 2. Drills of 1/2 inch capacity shall have a breastplate and spade end handle and two side handles, located at approximately the center of gravity of the drill, one of the side handles shall be removable and the other shall be stationary and contain a grip-controlled switch. The combination breastplate and spade end handles shall be either fixed horizontal (Figure 4), fixed vertical (Figure 5), or be of a removable type (Figure 6) which permits adjusting from the horizontal to vertical positions.

Source: https://assist.dla.mil -- Downloaded: 202610-26T11:33Z
Check the source to verify that this is the current version before use.

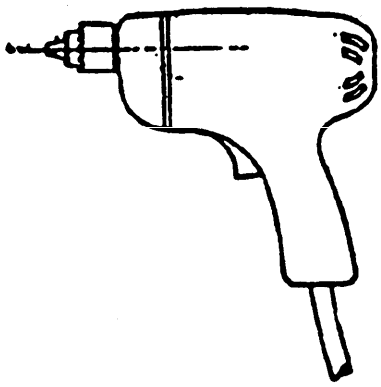


Figure 3. End handle pistol grip (1/4 inch and 3/8 inch capacity drills).

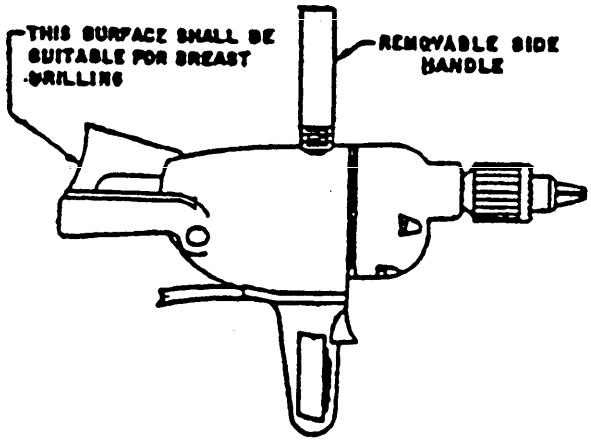


Figure 4. Combination breastplate and spade handle nonremovable horizontal type (3/8 inch and 1/2 inch capacity drills).

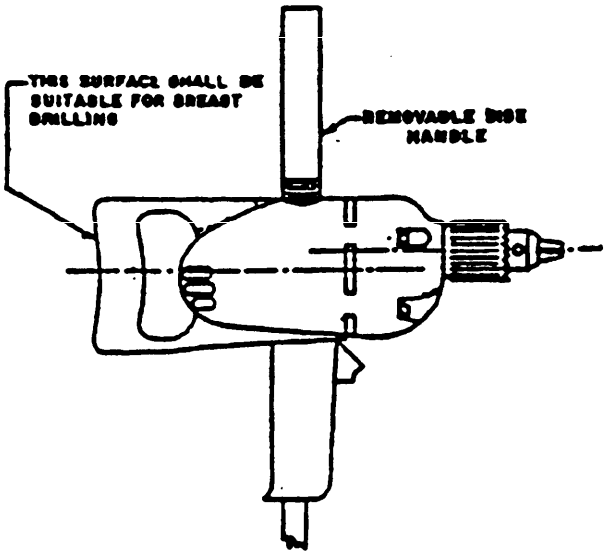


Figure 5. Combination breastplate and spade handle nonremovable vertical type (3/8 inch and 1/2 inch capacity drills).

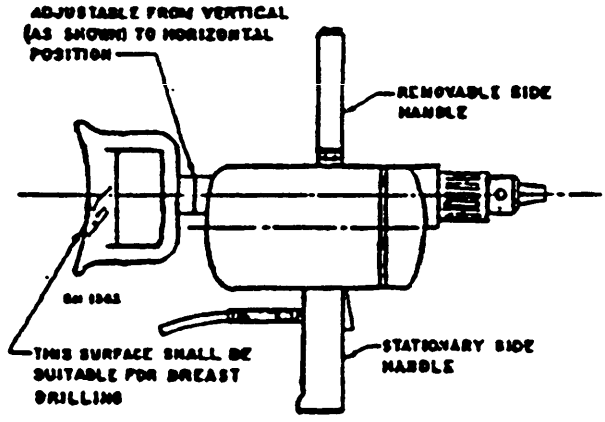


Figure 6. Combination breastplate and spade handle removable type. Stationary and removable side handles (1/2 inch capacity drills and over).

3.7.3 Type III drills. The type III drills shall be suitable for general purpose drilling of mild steel. The type III drills shall conform to Table III requirements for the class, size, and styles specified (see 6.2). When specified (see 3.6.2 and 6.2), type III, class A, size 1/2 to 1-1/4 inch capacity drills shall be capable of reversible operation.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:33Z
Check the source to verify that this is the current version before use.

Table III. Type III drills

Size (inch)	Chuck capacity (min.) (inch)	Weight without cord (max.) (pounds)	Length overall (max.) (inches)	Spindle revolutions per minute		Rated load amps at 120 volts AC (min.)	Torque at spindle shaft (min.)	
				No load (max.)	Rated load 1/		Rated load (foot-pounds)	Stall (foot-pounds)
Class A, heavy duty								
1/4	0 - 1/4	8.5	15	2500	1000 - 1700	2.5	1.0	5.0
3/8	0 - 3/8	15.0	18	1200	400 - 660	3.5	3.0	15.0
1/2	5/64 - 1/2	21.0	20	650	200 - 400	6.0	4.7	27.5
3/4	3/16 - 3/4	25.0	21.5	450	200 - 350	10.0	10.0	62.0
1	- -	34.0	25	420	180 - 235	9.0	17.0	80.0
1-1/4	- -	66.0	25	350	130 - 200	11.0	33.0	165.0
Class B, style 1, straight drive, standard duty								
1/4	0 - 1/4	6.5	12.0	2500	1000 - 1600	1.8	0.56	3.0
3/8	0 - 3/8	10.5	15.5	1200	450 - 660	2.5	2.0	10.0
1/2	0 - 1/2	12.0	15.0	650	200 - 400	3.0	4.0	25.0
Class B, style 2, - 90° angle drive								
1/4	0 - 1/4	4.0	10.25	2500	1000 - 1600	1.3	0.3	1.5
Class B, style 3 - 55° angle drive								
1/4	0 - 1/4	4.0	9.0	2500	950 - 1500	1.3	0.3	1.5

1/ The rated load speed shall in no case be less than 55 percent of the rated no load speed.

3.7.3.1 Handles. The type III, class A drills of 1/4 inch and 3/8 inch capacity shall have an end handle with a pistol grip similar to Figures 1 or 3, or an end handle with a closed grip similar to Figure 2, as specified (see 6.2). When specified (see 6.2), 3/8 inch drills shall be supplied with an end handle with a closed grip and an auxiliary side handle (see Figure 7). Unless otherwise specified (see 6.2), drills of 3/8 inch through 1-1/4 inch capacity shall have a combination breastplate and spade handle of the removable type similar to Figure 6, fixed horizontal similar to Figure 4 or fixed vertical similar to Figure 5, and two side handles; one side handle to be removable and the other stationary with a grip-controlled switch. When specified (see 6.2), drills of 1/2 inch capacity and larger shall have a removable type combination breastplate and spade handle similar to Figure 6 which permits adjustment from the horizontal to vertical positions, or fixed horizontal similar to Figure 4 or fixed vertical similar to Figure 5.

Source: https://assist.dla.mil -- Downloaded: 202640-26T11:33Z
Check the source to verify that this is the currentversion before use.

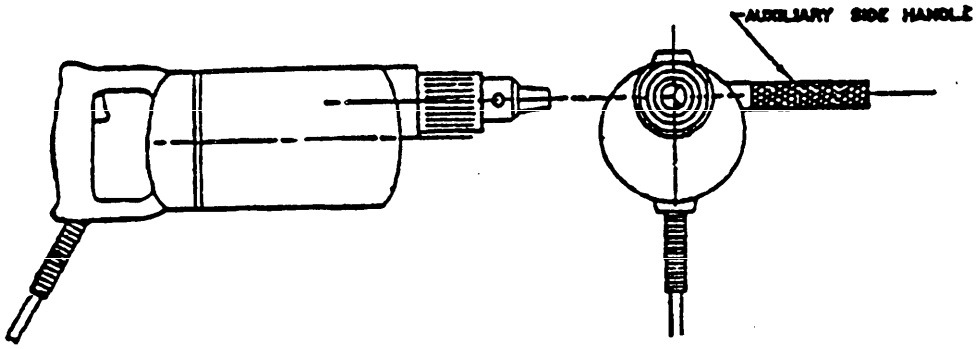


Figure 7. End handle with closed grip and an auxiliary side handle (3/8 inch capacity drill).

3.7.3.1 Feed screws. When specified (see 6.2), the type III, class A drills of 1/2 inch capacity and greater shall be equipped with a screw feed attachment similar to Figure 8. The feed screws shall provide at least a 2-1/2 inch feed and shall be made of steel with four radiating arms or continuous rim with suitable spokes forehand-feed control. Feed screw construction shall be such that there will be no unsteadiness when the drill is operated with the feed screw in any position up to maximum extension. The outer or external end of the feed screw shall have a hardened steel center.

3.7.3.2 Type III, class B standard duty. The class B standard duty drills shall conform to the requirements of Table III for class B standard duty drills for the size specified (see 6.2).

3.7.3.2.1 Class B drill handles. The class B drill handles shall conform to the handle requirements for class A for applicable size drills, except drills of 1/2 inch capacity may be furnished with a fixed breastplate and spade handle similar to Figures 4 and 5.

3.7.3.2.2 Class B, style 1. The style 1 drills shall have an end handle with a pistol grip. The style 1 drills shall be similar to Figures 1 or 3 and conform to the requirements of Table III for class B, style 1 drills.

3.7.3.2.3 Class B, style 2, 90° drive. Style 2 drills shall have a body type handle with the chuck spindle at a 90° angle to the body. A nonlocking instant release type switch shall be provided. Class B, style 2 drills shall be similar to Figure 9 and shall conform to the requirements of Table III for class B, style 2, 90° angle drills.

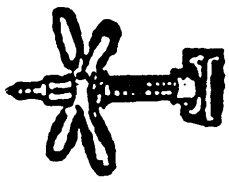


Figure 8. Feed screw.

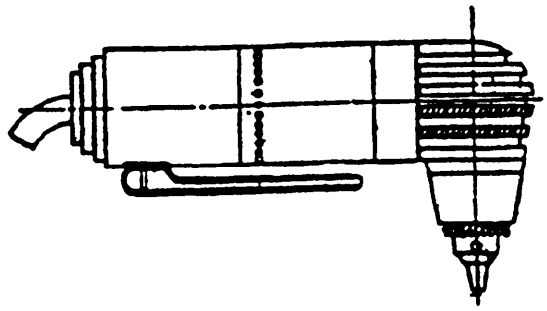


Figure 9. Class B, style 2, 90° handle driver, 1/4 inch capacity (body handle and instant release switch).

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:33Z
Check the source to verify that this is the current version before use.

3.7.3.2.4 Class B, style 3, 55° handle drive. The style 3 drills shall have a body type handle with the chuck spindle at approximately 55° to the body. A nonlocking instant release type switch shall be provided. The class B, style 3 drills shall be similar to Figure 10 and shall conform to the requirements of Table III for class B, style 3, 55° handle drills.

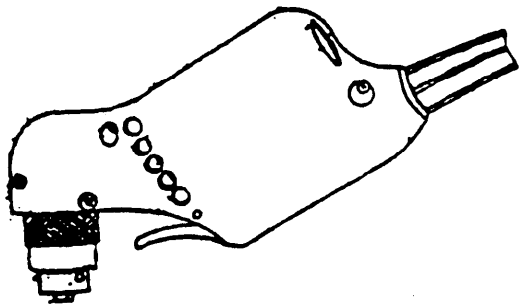


Figure 10. Class B, style 3, 55° handle drive 1/4 inch capacity (body handle and instant release switch).

3.8 Type IV drill stands with drills. The type IV drill stands with drills furnished under this specification shall consist of a vertical adjustable stand with a portable electric drill for use as a bench type drilling machine. Electric drills shall comply with type III, class A, heavy duty drills, as specified herein, and shall be designed for mounting on the stand. A means shall be provided for holding the switch in the "ON" position. Drill stands with drills shall be similar to Figure 11 and comply with the requirements of Table IV for the size specified (see 6.2).

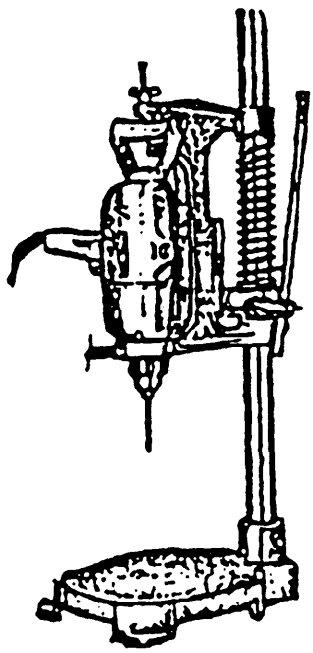


Figure 11. Type IV drill stand with drill.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:33Z
Check the source to verify that this is the current version before use.

Table IV. Type IV drill stands with drills

Size number (inches)	Electric drill, size type III, class A (inches)	"Capacity Swing" face of column to center of chuck (min.) (inches)	"Feed" travel of drill in clamped position (at least) (inches)	Vertical adjustment on column (max. height drill chuck to base) (at least) (inches)	Weight complete with drill (max.) (inches)	Working surface of base size (approx.) (square) (inches)
1/4	1/4	4-3/4	2	10	35	30
3/8	3/8	5-1/2	4	12	65	50
1/2	1/2	4-5/8	2-1/2	12	65	50
3/4	3/4	5	4	12	80	80
1	1	6	4	12	85	80
1-1/4	1-1/4	11	4	16	125	100

3.8.1 Stand. The drill stand shall be designed for holding an electric drill of the size specified (see 6.2). The stand shall consist of a base, column, drill mounting fixture, and operating lever. The base shall be a casting with top machined flat as a table working surface and provided with at least two integral lugs with holes for bench mounting. The base shall be provided with a means for mounting and rigidly holding a round column. The column shall be of a diameter to fully support the drill in operation and shall be accurately machined. The column shall be of sufficient height to provide the vertical adjustment specified in Table IV for the size specified. The drill mounting fixture shall be designed for securely holding the drill with mounting brackets fitted to and mounted on the column. A column clamp, spring, and operating level providing feed shall be included in the design. The column clamp shall be fitted to the column and be adjustable to any position within the adjustment range. The spring shall be heavy enough to immediately return the drill to the top position when the lever is released. The lever shall provide a leverage ratio of at least 3-1/2 to 1.

3.9 Electromagnetic interference suppression. When specified (see 6.2), each drill shall be electromagnetic interference free within the limits specified for portable electric tools in compliance with the requirements of MIL-STD-461.

3.10 Moisture and fungus treatment. When specified (see 6.2), the electrical components of drills covered by this specification shall be moisture- and fungus-resistant treated with material conforming to MIL-V-173 with the following exceptions:

(a) Components or circuit elements which are inherently fungus- and water-resistant or which are hermetically sealed need not be treated.

(b) Components or circuit elements whose function will be adversely affected by the varnish coating shall not be treated.

When used, the varnish shall be applied to give a minimum dry film thickness of 0.001 inch to the component or element surface previously cleaned and prepared in accordance with accepted commercial practice.

3.11 Identification plate. Each drill shall bear an identification plate in accordance with the applicable requirements of UL 45.

3.12 Handbook of instructions. Unless otherwise specified (see 6.2), each drill shall be furnished with an instruction manual in accordance with the applicable requirements of UL 45.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:33Z
Check the source to verify that this is the current version before use.

3.13 Tools and extra parts. Drills with a geared chuck shall be provided with a chuck key and thong or a key holder attached to the cord. Each drill shall be furnished with maintenance tools and extra parts that are normally supplied commercially.

3.14 Metric products. Products manufactured to metric dimensions will be considered on an equal basis with those manufactured using inch-pound units, provided they fall within the tolerances specified and all other requirements of this document are met. If a product is manufactured to metric dimensions and those dimensions exceed the tolerances specified in the inch-pound units, a request should be made to the specification preparing activity for change to this document.

3.15 Workmanship. Details of workmanship shall be in accordance with the best commercial standards and practices. Paints, coatings, platings, and finishes shall be smooth, adherent, continuous, and not stained or discolored. Fasteners shall be firmly secured and show no evidence of deformation, cross threading, or hazardous burrs. Adhesives and lubricants adequate for their intended purpose shall be properly and neatly applied. Adhesives shall be adequately cured. Wires and cables shall be neatly dressed and shall not be frayed or in contact with sharp edges. Wire and cable insulation shall show no evidence of burns, abrading, or pinch marks. There shall be to interference, binding, or galling of parts. External and bearing surfaces shall be free of tool and gouge marks, nicks, or other surface imperfections. The item shall be clean and free of corrosion and debris (e.g., chips, shavings, slivers) or other foreign material. The item shall be free from manufacturing workmanship defects, (e.g., loose, missing, binding, misaligned parts, sharp or rough external edges, corners, or surfaces) and material workmanship defects (e.g., pits, rips, fins, burrs, tears, nodules, cracks, blisters) which may adversely impact the item's serviceability, durability, safety, or appearance.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.1.1 Component and material inspection. In accordance with 4.1 above, the supplier is responsible for insuring that components and materials used were manufactured, tested, and inspected in accordance with the requirements of referenced subsidiary specification and standards to the extent specified, or, if none, in accordance with this specification.

4.2 Classification of inspection. Inspection shall be classified as follows:

(a) First article inspection.
(b) Quality conformance inspection.

4.2.1 First article inspection. First article inspection shall be applied to the first article submitted in accordance with 3.2. Failure of the first article to pass the examination on one or more preproduction tests shall be cause for rejection. Unless otherwise specified (see 6.2), first article inspection shall consist of the examination in 4.4.1 and all tests in 4.5.

4.3 Sampling for quality conformance inspection.

4.3.1 Lot. All drills of the same type, class, and style presented at one time shall be considered a lot for purposes of quality conformance inspection.

4.3.2 Sample. A random sample shall be selected from each lot in conformance with ANSI/ASQC Z1.4

4.4 Quality conformance inspection. Quality conformance inspection shall be performed on each sample drill selected in accordance with 4.3.2. Quality conformance inspection shall consist of the following:

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:33Z
Check the source to verify that this is the current version before use.

(a) Examination (see 4.4.1).
(b) Tests (see 4.5.1, 4.5.2, and 4.5.3).
(c) Inspection of preparation for delivery requirements (see 4.6).

4.4.1 Visual examination. The end items shall be examined for the defects listed in Table V. The lot size shall be expressed in terms of drills. The inspection level shall be I and the acceptable quality level (AQL), expressed in terms of percent defective, shall be 2.5 for major and 6.5 for total (major and minor combined) defects.

4.4.2 Dimensional examination. The end items shall be examined for any nonconformance to dimensional requirements. The lot size shall be expressed in terms of drills. The sample unit shall be one drill. The inspection level shall be I and the AQL, expressed in terms of percent defective, shall be 6.5.

Table V. Classification of defects

Examine	Defect	Classification		
		Critical	Major	Minor
Construction and workmanship	Evidence of electric shock or fire hazard	001		
	Drills not of type, class, or style specified		101	
	Style of handle not as specified		102	
	Component parts missing, damaged, defective, or not as specified		103	
	Cable assembly not length specified, not integrally connected. Strain relief at cord entry into tool not suitable - allows strain on internal wiring. Attachment plug defective, not secure on cord, prongs improper size, loose or broken		104	
	Brush caps (if used) not properly insulated or protected from mechanical abuse		105	
	Switch defective, loose, or binds, not rigidly mounted or no provision for holding in "ON" position, as applicable		106	
	Evidence of lubricant leakage from gear housing or bearings		107	
	Chuck not type specified, key or thong missing		108	
	Chuck defective, fails to firmly and centrally hold straight shank drills of required sizes		109	
	Drill stand with drill, feed action irregular, operation not smooth, spring of insufficient strength to return and hold drill and holding fixture in raised position		110	
	Dimensions and weight not as specified			201
	Identification, nameplate does not conform, illegible or not permanent			202
	Improper workmanship, not smooth finished			203
	Instruction book and parts lists missing, data not as specified			204
	Drill stands size, capacity, weight, and dimensions not as specified			205

4.4.3 Tests for quality conformance. Each sample selected in accordance with 4.3.2 shall be tested for compliance with the requirements of this specification. The AQL shall be 2.5 percent defective. Nonconforming drills shall not be offered for delivery and if the number of defective units exceeds the acceptable number for the sample tested, the lot represented shall be rejected. Quality conformance tests shall consist of the tests specified in 4.5.1, 4.5.2, and 4.5.3.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:33Z
Check the source to verify that this is the current version before use.

4.5 Tests.

4.5.1 Electromagnetic interference test certification. If an EMI suppression network is required, the contractor shall provide certification that the suppression network in the drills offered conform to the requirements of 3.9, or are identical to that of drills previously tested and approved.

4.5.2 UL 45 tests certification. The offerer shall provide certification that the drill offered conforms to the test requirements of UL 45 (see 3.4.1 for compliance to this requirement).

4.5.3 Speed, amperage, and other tests. Sample drills shall be tested to verify that the drills comply with the requirements specified below and as listed on the nameplate. Variation in values shall not be more than percentages allowed by UL 45. Nonconforming drills shall be rejected:

(a) No-load and rated revolutions per minute.
(b) Rated load amps at nameplate voltage.
(c) Rated load and stalled torque at spindle shaft.

4.6 Inspection of preparation for delivery requirements. An inspection shall be made to determine that the preservation, packaging, packing, and marking comply with the requirements in section 5 of this specification. Defects shall be scored in accordance with Table VI. The lot size shall be the number of shipping containers in the end item inspection lot. The inspection level shall be S-2 and the AQL, expressed in terms of defects per hundred units, shall be 4.0.

Table VI. Classification of preparation for delivery defects

Examine	Defect
Marking (exterior and interior)	Omitted; incorrect; illegible; improper size, location, sequence, or method of application
Materials	Any component missing, damaged, or not as specified
Workmanship	Inadequate application of components such as incomplete closure of container flaps, loose strapping, inadequate stapling Distortion of container
Contents	Number per container is more or less than required Net weight exceeds requirements

5. PREPARATION FOR DELIVERY

5.1 Preservation, packaging, and packing. Preservation, packaging, and packing shall be in accordance with PPP-P-40. The level of preservation, packaging, and packing shall be A, B, or Commercial, as specified (see 6.2).

5.2 Marking.

5.2.1 Civil agencies. In addition to any special marking specified in the contract or order, each unit package and shipping container shall be marked in accordance with FED-STD-123.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:33Z
Check the source to verify that this is the current version before use.

5.2.2 Military agencies. Interior packages and shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful but is not mandatory.)

6.1 Intended use. The electric drills covered by this specification are compact portable tools with universal motors for operation using 115 volt DC and 115 volts 25 to 60 cycles AC and designed for drilling or reaming metal or wood more or less intermittently according to type. The type II drills are used for high speed intermittent drilling of aluminum and nonferrous metals of equivalent hardness. The type II drills are used for heavy duty drilling of corrosion-resistant steels. The type III, class A drills are used for continuous heavy duty drilling of mild steel. The type III, class B, drills are used for light duty drilling and intermittent heavy duty drilling.

6.2 Ordering data. Acquisition documents must specify the following:

a. Title, number, and date of this specification.
b. Type, class, and style, if applicable, of item required (see 1.2).
c. First article inspection, when required (see 3.2, 4.2.1, and 6.3).
d. Specify whether drill to be double insulated or 3-wire grounded (see 3.4).
e. Specify cord length required (see 3.6.3).
f. Chuck or taper requirements if different (see 3.6.9).
g. When 3/4 inch to 1-1/4 inch capacity drills are to be reversible (see 3.7.3).
h. Handle required if different (see 3.7.1.1 and 3.7.3.1).
i. When type III, class A drills of 1/2 inch capacity or greater are to be furnished with feed screw attachment (see 3.7.3.1).
j. Size of drill stand and drill for type IV (see 3.8 and 3.8.1).
k. Specify electromagnetic interference suppression when required (see 3.9).
l. Moisture and fungus treatment when required (see 3.10).
m. Handbook of instructions if different (see 3.12).
n. Switch rating if other than specified (see 3.6.2).
o. Level of preservation, packaging, and packing required (see 5.1).
p. Marking, if other than specified.

MILITARY INTERESTS:

Army - GL
Navy - SH
Air Force - 84

MILITARY COORDINATING ACTIVITY:

Army - GL

Preparing Activity:

GSA - FSS

Review Activities:

Army - RL, GL
Navy - SH
Air Force - 84, 85
DSAH - GS

User Activities:

Army - CE, ME
Navy - AS, MC, OS, YD, CG

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:33Z
Check the source to verify that this is the current version before use.